Proof Clearly the result follows provided

$$\sum_{j=1}^{\mathrm{N}_c} \mathrm{d}q_j \wedge \mathrm{d}\left[\mu \nabla \gamma(\boldsymbol{q})\right]_j = 0,$$

where $[\ldots]_j$ indicates the jth component of the term in brackets. We have

$$\sum_{j=1}^{\mathrm{N}_c} \mathrm{d}q_j \wedge \mathrm{d}\left[\mu \nabla \gamma(\boldsymbol{q})\right]_j = \sum_{j=1}^{\mathrm{N}_c} \mathrm{d}q_j \wedge \left(\frac{\partial \gamma}{\partial q_j} \mathrm{d}\mu + \mu \sum_{i=1}^{\mathrm{N}_c} \gamma''_{ij} \mathrm{d}q_j \right).$$

Then we note

$$\sum_{j=1}^{\mathrm{N}_c} \mathrm{d}q_j \wedge \frac{\partial \gamma}{\partial q_j} \mathrm{d}\mu = \sum_{j=1}^{\mathrm{N}_c} \frac{\partial \gamma}{\partial q_j} \mathrm{d}q_j \wedge \mathrm{d}\mu = 0,$$

since $\gamma(q) \equiv 0$ implies $\mathrm{d}\gamma(q) = \sum_{j=1}^{\mathrm{N}_c} \frac{\partial \gamma}{\partial q_j} \mathrm{d}q_j \equiv 0$. Moreover

$$\sum_{j=1}^{\mathrm{N}_c} \mathrm{d}q_j \wedge \sum_{i=1}^{\mathrm{N}_c} \gamma''_{ij} \mathrm{d}q_j = 0,$$

since the Hessian matrix γ'' is symmetric. □

Observe that

$$\boldsymbol{g}'(\boldsymbol{q}_{n+1})^T \boldsymbol{\mu}_{n+1} = \sum_{k=1}^{l} \mu_{n+1,k} \nabla g_k(\boldsymbol{q}_{n+1}).$$

Thus we may apply Lemma 4.1 with γ replaced by each g_k to demonstrate

$$\sum_{j=1}^{\mathrm{N}_c} \mathrm{d}q_{n+1,j} \wedge \mathrm{d}p_{n+1,j} = \sum_{j=1}^{\mathrm{N}_c} \mathrm{d}q_{n+1,j} \wedge \mathrm{d}\bar{p}_{n+1,j}.$$

Using similar argumentation based on the properties of the wedge product, we next show

$$\sum_{j=1}^{\mathrm{N}_c} \mathrm{d}q_{n+1,j} \wedge \mathrm{d}\bar{p}_{n+1,j} = \sum_{j=1}^{\mathrm{N}_c} \mathrm{d}q_{n,j} \wedge \mathrm{d}\bar{p}_{n+1,j},$$

and, finally, invoking Lemma 4.1 again,

$$\sum_{j=1}^{\mathrm{N}_c} \mathrm{d}q_{n,j} \wedge \mathrm{d}\bar{p}_{n+1,j} = \sum_{j=1}^{\mathrm{N}_c} \mathrm{d}q_{n,j} \wedge \mathrm{d}p_{n,j}.$$